

Global and local cohomological invariants relative to a closed subspace

SGA 2, Exposé I

English draft

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About this draft

This is an unofficial English translation of Exposé I, “Global and local cohomological invariants relative to a closed subspace”, from SGA 2. The source volume is A. Grothendieck (notes by a group of auditors), with an exposé by M. Raynaud, *Cohomologie locale des faisceaux cohérents et théorèmes de Lefschetz locaux et globaux* (SGA 2), Séminaire de géométrie algébrique du Bois Marie, 1962. It follows the slightly corrected SMF recomposition, arXiv:math/0511279.

This draft contains the assigned body of the source, including proofs, original footnotes, and editor notes (N.D.E.). Statement numbering and mathematical notation follow the source. References to material outside this draft retain their original numbers. Apparent mathematical misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ I

Global and local cohomological invariants relative to a closed subspace

1. The functors $\Gamma_Z, \underline{\Gamma}_Z$

Let X be a topological space, $\underline{\mathcal{C}}_X$ the category of abelian sheaves on X . Let Φ be a family of supports in the sense of Cartan; one defines the functor Γ_Φ on $\underline{\mathcal{C}}_X$ by:

- (1) $\Gamma_\Phi(F) =$ subgroup of $\Gamma(F)$ consisting of the sections f such that support $f \in \Phi$.

If Z is a closed subset of X , we denote by abuse of language by $\underline{\Gamma}_Z$ the functor Γ_Φ , where Φ is the set of closed subsets of X contained in Z . Thus one has:

- (2) $\Gamma_Z(F) =$ subgroup of $\Gamma(F)$ consisting of the sections f such that support $f \subset Z$.

We wish to generalize this definition to the case where Z is a *locally closed* subset of X , hence closed in a suitable open subset V of X . In this case one sets:

- (3)
$$\Gamma_Z(F) = \Gamma_Z(F|_V).$$

One must check that $\Gamma_Z(F)$ “does not depend” on the chosen open set. It suffices to show that if $V', V \supset V' \supset Z$ is an open set, then the map $\rho_{V'}^V: F(V) \rightarrow F(V')$ maps $\Gamma_Z(F|_V)$ isomorphically onto $\Gamma_Z(F|_{V'})$. Now

- (4)
$$\Gamma_Z(F|_V) = \ker \rho_{V-Z}^V$$

hence if $f \in \Gamma_Z(F|_V)$ and if $\rho_{V'}^V(f) = \rho_{V-Z}^V(f) = 0$ then $f = 0$ since $(V', V - Z)$ is a covering of V . Likewise, if $f' \in \Gamma_Z(F|_{V'})$, then $f' \in F(V')$ and $0 \in F(V - Z)$ define an $f \in F(V)$ such that $\rho_{V'}^V(f) = f'$, $f \in \Gamma_Z(F|_V)$, hence $\rho_{V'}^V$ induces an isomorphism $\Gamma_Z(F|_V) \rightarrow \Gamma_Z(F|_{V'})$.

Note that every open set W of Z is induced by an open set U of X in which W is closed. It follows that $W \mapsto \Gamma_W(F)$ defines a presheaf on Z , and one checks that it is a sheaf, which will be denoted $i^!(F)$, where $i: Z \rightarrow X$ is the canonical immersion. One finds:

- (5)
$$\Gamma_Z(F) = \Gamma(i^!(F)).$$

The sheaf $i^!(F)$ is a subsheaf of $i^*(F)$; indeed the canonical homomorphism:

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

is injective on $\Gamma_{U \cap Z}(F|_U) \subset \Gamma(F|_U)$. Summarizing, one has the following result:

PROPOSITION 1.1. *There exists a unique subsheaf $i^!(F)$ of $i^*(F)$ such that for every open set U of X such that $U \cap Z$ is closed in U ,*

$$\Gamma(F|_U) = \Gamma(U, F) \longrightarrow \Gamma(U \cap Z, i^*(F))$$

induces an isomorphism $\Gamma_{U \cap Z}(F|_U) \rightarrow \Gamma(U \cap Z, i^!(F))$.

Note that if Z is an open set one will simply have

- (6)
$$i^!(F) = i^*(F) = F|_Z, \quad \Gamma_Z(F) = \Gamma(Z, F).$$

Suppose again that Z is arbitrary. Then for a variable open set U of X , one sees that

$$U \longmapsto \Gamma_{U \cap Z}(F|_U) = \Gamma(U \cap Z, i^!(F))$$

is a sheaf on X , which we shall denote $\underline{\Gamma}_Z(F)$; more precisely, according to the preceding formula (expressing that $i^!$ commutes with restriction to open sets) one has an isomorphism

$$(7) \quad \underline{\Gamma}_Z(F) = i_*(i^!(F))$$

by definition, one has, for every open set U of X ,

$$(8) \quad \Gamma(U, \underline{\Gamma}_Z(F)) = \Gamma_{U \cap Z}(F|_U).$$

Let us note here a characteristic difference between the case where Z is closed, and that where Z is open. In the first case, formula (8) shows us that $\underline{\Gamma}_Z(F)$ may be regarded as a subsheaf of F , and one therefore has a *canonical immersion*

$$(8') \quad \underline{\Gamma}_Z(F) \hookrightarrow F.$$

In the case where Z is open, on the contrary, one sees from (6) that the second member of (8) is $\Gamma(U \cap Z, F)$, hence receives $\Gamma(U, F)$, hence one has a *canonical homomorphism* in the opposite direction from the preceding one:

$$(8'') \quad F \longrightarrow \underline{\Gamma}_Z(F),^1$$

which is moreover nothing other than the canonical homomorphism

$$F \longrightarrow i_*i^*(F),$$

taking into account the isomorphism

$$(6 \text{ bis}) \quad \underline{\Gamma}_Z(F) \simeq i_*i^*(F)$$

deduced from (6) and (7).

Of course, for variable F , $\Gamma_Z(F)$, $\underline{\Gamma}_Z(F)$, $i^!(F)$ may be regarded as functors in F , taking values respectively in the category of abelian groups, of abelian sheaves on X , of abelian sheaves on Z . It is sometimes convenient to interpret the functor

$$i^!: \underline{\mathbb{C}}_X \longrightarrow \underline{\mathbb{C}}_Z$$

as the adjoint functor of a well-known functor

$$i!: \underline{\mathbb{C}}_Z \longrightarrow \underline{\mathbb{C}}_X$$

defined by the following proposition:

PROPOSITION 1.2. *Let G be an abelian sheaf on Z . Then there exists a unique subsheaf of $i_*(G)$, namely $i_!(G)$, such that for every open set U of X the isomorphism (identical)*

$$\Gamma(U \cap Z, G) = \Gamma(U, i_*(G))$$

defines an isomorphism

$$\Gamma_{\Phi_{U \cap Z, U}}(U \cap Z, G) = \Gamma(U, i_!(G)),$$

where $\Phi_{U \cap Z, U}$ denotes the set of subsets of $U \cap Z$ which are closed in U .

The verification reduces to noting that the first member is a sheaf for variable U , i.e. that the property for a section of $i_*(G)$ over U , regarded as a section of G over $U \cap Z$, of having closed support *in U* is of *local nature on U* . The sheaf $i_!(G)$ just defined is also known under the name: *sheaf deduced from G by extending by 0 outside of Z* , cf. [Godement]. In particular, if Z is closed, one has

$$(9) \quad i_!(G) = i_*(G);$$

but in the general case, the canonical injection $i_!(G) \rightarrow i_*(G)$ is not an isomorphism, as is already well known for Z open. Obviously, $i_!(G)$ depends functorially on G (and it is even an exact functor in G). This being said, one has:

¹**N.D.E.** the original reference was (8).

PROPOSITION 1.3. *There exists an isomorphism of bifunctors in G, F (G an abelian sheaf on Z , F an abelian sheaf on X):*

$$(10) \quad \text{Hom}(i_!(G), F) = \text{Hom}(G, i^!(F)).$$

To define such an isomorphism, it amounts to the same to define functorial homomorphisms

$$i_!i^!(F) \longrightarrow F, \quad G \longrightarrow i^!i_!(G),$$

satisfying the well-known compatibility conditions (cf. for example Shih's exposé in the Cartan seminar on cohomological operations).

Recalling that $i_!$ is exact, hence transforms monomorphisms into monomorphisms, one concludes:

COROLLARY 1.4. *If F is injective, $i^!(F)$ is injective, hence $\underline{\Gamma}_Z(F) = i_*i^!(F)$ is likewise injective.*

Replacing X by a variable open set U of X , one also concludes from 1.3

COROLLARY 1.5. *One has a functorial isomorphism in F, G :*

$$(11) \quad \underline{\text{Hom}}(i_!(G), F) = i_*(\underline{\text{Hom}}(G, i^!(F))).$$

Taking for G the constant sheaf on Z defined by $\mathbf{Z}$, namely $\mathbf{Z}_Z$, 1.3 and 1.5 specialize to

COROLLARY 1.6. *One has functorial isomorphisms in F :*

$$(12) \quad \begin{aligned} \Gamma_Z(F) &= \text{Hom}(\mathbf{Z}_{Z,X}, F), \\ \underline{\Gamma}_Z(F) &= \underline{\text{Hom}}(\mathbf{Z}_{Z,X}, F), \end{aligned}$$

where $\mathbf{Z}_{Z,X} = i_!(\mathbf{Z}_Z)$ is the abelian sheaf on X deduced from the constant sheaf on Z defined by $\mathbf{Z}$, by extending by 0 outside of Z .

REMARK 1.7. Suppose that X is a ringed space, and endow Z with the sheaf of rings $\mathcal{O}_Z = i^{-1}(\mathcal{O}_X)$, and finally denote by $\underline{\mathbf{C}}_X$ and $\underline{\mathbf{C}}_Z$ the category of Modules on X resp. Z . Then the preceding considerations extend word for word, taking for F a Module on X and for G a modules on Z , and interpreting accordingly the statements 1.3 to 1.6.

To finish these generalities, let us examine what happens when one changes the locally closed subset Z . Let $Z' \subset Z$ be another locally closed subset, and let

$$j: Z' \longrightarrow Z, i': Z' \longrightarrow X, i' = ij$$

be the canonical inclusions. Then one has functorial isomorphisms:

$$(13) \quad (ij)^! = j^!i^!, \quad (ij)_! = i_!j_!.$$

The first isomorphism (13) defines a functorial isomorphism

$$(14) \quad \Gamma_{Z'}(F) = \Gamma(Z', (ij)^!(F)) \simeq \Gamma(Z', j^!(i^!(F))) = \Gamma_{Z'}(i^!(F)).$$

Suppose now that Z' is *closed* in Z , and

$$Z'' = Z - Z'$$

its complement in Z , which is open in Z hence locally closed in X . The canonical inclusion (8') applied to $i^!(F)$ on Z equipped with Z' defines for us, thanks to (14), a canonical injective functorial homomorphism

$$(15) \quad \Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F).$$

If one replaces in (14) Z' by Z'' and uses (8''), one finds a canonical functorial homomorphism:

$$(15') \quad \Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F).$$

PROPOSITION 1.8. *Under the preceding conditions, the sequence of functorial homomorphisms:*

$$(16) \quad 0 \longrightarrow \Gamma_{Z'}(F) \longrightarrow \Gamma_Z(F) \longrightarrow \Gamma_{Z''}(F)$$

is exact. If F is flasque, the sequence remains exact upon putting a zero on the right.

PROOF. Replacing X by an open set V in which Z is closed, one is reduced to the case where Z is closed, hence Z' closed. Then Z'' is closed in the open set $X - Z'$, and one has a canonical inclusion

$$\Gamma_{Z''}(F) \longrightarrow \Gamma(X - Z', F),$$

and the exactness of (16) simply means that the sections of F with support in Z' are those whose restriction to $X - Z'$ is zero.

When F is flasque, every element of $\Gamma_{Z''}(F)$, regarded as a section of F over $X - Z'$, can be extended to a section of F over X , and the latter will evidently have its support in Z , which proves that then the last homomorphism in (16) is surjective. $\square$

COROLLARY 1.9. *One has a functorial exact sequence*

$$(16 \text{ bis}) \quad 0 \longrightarrow \underline{\Gamma}_{Z'}(F) \longrightarrow \underline{\Gamma}_Z(F) \longrightarrow \underline{\Gamma}_{Z''}(F),$$

and if F is flasque, this sequence remains exact upon putting a 0 on the right.

One can interpret (1.8) in terms of results on the functors Hom and $\underline{\text{Hom}}$ via 1.6, in the following way. Let us first note that if G is an abelian sheaf on Z , inducing the sheaves $j^*(G)$ and $k^*(G)$ on Z' resp. Z'' (where $j: Z' \rightarrow Z$ and $k: Z'' \rightarrow Z$ are the canonical injections), one has a canonical exact sequence of sheaves on X :

$$(17) \quad 0 \longrightarrow k^*(G)_X \longrightarrow G_X \longrightarrow j^*(G)_X \longrightarrow 0$$

where to simplify the notations, the index X denotes the sheaf on X obtained by extending by 0 in the complement of the space of definition of the sheaf under consideration. The exact sequence (17) generalizes a well-known exact sequence when $Z = X$ (cf. [Godement]), and is moreover deduced from it by writing the exact sequence in question on Z , and applying the functor $i_!$. Taking $G = \mathbf{Z}_Z$, one concludes in particular:

PROPOSITION 1.10. *Under the preceding conditions, one has an exact sequence of abelian sheaves on X :*

$$(18) \quad 0 \longrightarrow \mathbf{Z}_{Z'', X} \longrightarrow \mathbf{Z}_{Z, X} \longrightarrow \mathbf{Z}_{Z', X} \longrightarrow 0.$$

This being granted, the two exact sequences 1.8 and 1.9 are nothing other than the exact sequences deduced from (18) by applying the functor $\text{Hom}(-, F)$ resp. $\underline{\text{Hom}}(-, F)$.

This gives again an evident proof of the fact that the sequences (16) and (16 bis) remain exact upon putting a zero on the right, provided that F is *injective*.

2. The functors $H_Z^*(X, F)$ and $\underline{H}_Z^*(F)$

DEFINITION 2.1. One denotes by $H_Z^*(X, F)$ and $\underline{H}_Z^*(F)$ the derived functors in F of the functors $\Gamma_Z(F)$ resp. $\underline{\Gamma}_Z(F)$.

These are cohomological functors, with values in the category of abelian groups resp. in the category of abelian sheaves on X . When Z is closed, $H_Z^*(X, F)$ is by definition none other than $H_\Phi^*(X, F)$ where Φ denotes the family of closed subsets of X contained in Z . When Z is open, one will see that $H_Z^*(X, F)$ is none other than $H^*(Z, F) = H^*(Z, F|_Z)$, thanks to the following proposition.

PROPOSITION 2.2 (Excision theorem). *Let V be an open subset of X containing Z . Then one has an isomorphism of cohomological functors in F :*

$$(19) \quad H_Z^*(X, F) \longrightarrow H_Z^*(V, F|_V).$$

Indeed, one has a functorial isomorphism $\Gamma_Z^X \simeq \Gamma_Z^V j^!$, where $j : V \rightarrow X$ is the inclusion and where $j^!$ is therefore the restriction functor (cf. (14)). The latter is exact, and transforms injectives into injectives by 1.4, whence immediately the isomorphism (19).

When Z is open, one may take $V = Z$ and one finds:

COROLLARY 2.3. *Suppose Z is open, then one has an isomorphism of cohomological functors:*

$$(20) \quad H_Z^*(X, F) = H^*(Z, F).$$

One concludes from the isomorphisms 1.6 and from the definitions (cf. [Tôhoku]):

PROPOSITION 2.3 BIS². *One has isomorphisms of cohomological functors:*

$$(21) \quad H_Z^*(X, F) \simeq \text{Ext}^*(X; \mathbf{Z}_{Z,X}, F),$$

$$(21 \text{ bis}) \quad \underline{H}_Z^*(F) \simeq \underline{\text{Ext}}^*(\mathbf{Z}_{Z,X}, F).$$

One can therefore apply the results of [Tôhoku] on the Ext of Modules. Let us first point out the following interpretation of the sheaves $\underline{H}_Z^*(F)$ in terms of the global groups $H_Z^*(X, F)$:

COROLLARY 2.4. *$\underline{H}_Z^*(F)$ is canonically isomorphic to the sheaf associated to the presheaf*

$$U \longmapsto H_{Z \cap U}^*(U, F|_U).$$

In particular, using corollary 2.3, one finds:

COROLLARY 2.5. *Suppose Z is open, then one has an isomorphism of cohomological functors:*

$$(22) \quad \underline{H}_Z^*(F) = R^* i_* i^*(F)$$

(where $i : Z \rightarrow X$ is the inclusion).

The spectral sequence of the Ext gives the important spectral sequence:

THEOREM 2.6. *One has a spectral sequence functorial in F , abutting to $H_Z^*(X, F)$ and with initial term*

$$(23) \quad E_2^{p,q}(F) = H^p(X, \underline{H}_Z^q(F)).$$

REMARKS 2.7. It follows at once from 2.4 that the sheaves $\underline{H}_Z^q(F)$ are zero on $X - \overline{Z}$, and likewise zero in the interior of Z for $q \neq 0$ (hence for such a q , $\underline{H}_Z^q(F)$ is even supported on the boundary of Z).

Consequently, the second member of (23) may be interpreted as a cohomology group on $\overline{Z}$. We shall use 2.6 in the case where Z is closed in X , and where the second member of (23) may be interpreted as a cohomology group computed on Z :

$$(23 \text{ bis}) \quad E_2^{p,q}(F) = H^p(Z, \underline{H}_Z^q(F)).^3$$

Let us also note that when Z is open, the spectral sequence 2.6 is none other than the Leray spectral sequence for the continuous map $i : Z \rightarrow X$, taking into account the interpretation 2.5 in the computation of the initial term of the Leray spectral sequence.

Let us return to the exact sequence (18)⁴, it gives rise to an exact sequence of Ext (cf. [Tôhoku]):

²**N.D.E.** the proposition bears the number 2.3 in the original edition.

³**N.D.E.** the equation was numbered (23) in the original edition.

⁴**N.D.E.** the original reference was (1.10).

THEOREM 2.8. *Let Z be a locally closed subset of X , Z' a closed subset of Z and $Z'' = Z - Z'$. Then one has an exact sequence functorial in F :*

$$(24) \quad 0 \longrightarrow H_{Z'}^0(X, F) \longrightarrow H_Z^0(X, F) \longrightarrow H_{Z''}^0(X, F) \xrightarrow{\partial} H_{Z'}^1(X, F) \longrightarrow H_Z^1(X, F) \dots$$

$$\dots H_{Z'}^i(X, F) \longrightarrow H_Z^i(X, F) \longrightarrow H_{Z''}^i(X, F) \xrightarrow{\partial} H_{Z'}^{i+1}(X, F) \dots$$

Let us recall how one may obtain this exact sequence. Let $C(F)$ be an injective resolution of F , then the exact sequence (18)⁵ gives rise to the exact sequence

$$(25) \quad 0 \longrightarrow \Gamma_{Z'}(C(F)) \longrightarrow \Gamma(C(F)) \longrightarrow \Gamma_{Z''}(C(F)) \longrightarrow 0,$$

(which is none other than the one defined in 1.8). One concludes from this an exact sequence of cohomology, which is none other than (24).

The most important case for us is that where Z is closed (and moreover one can always reduce to this by replacing X by an open set V in which Z is closed). Then Z' is closed, Z'' is closed in the open set $X - Z'$, and one may write

$$(26) \quad H_{Z''}^i(X, F) = H_{Z''}^i(X - Z', F|_{X-Z'}),$$

which allows us to write the exact sequence (24) in terms of cohomologies with support in a given closed set. The most frequent case is that where $Z = X$. Setting then for simplicity $Z' = A$, one finds:

COROLLARY 2.9. *Let A be a closed subset of X . Then one has an exact sequence functorial in F :*

$$(27) \quad 0 \longrightarrow H_A^0(X, F) \longrightarrow H^0(X, F) \longrightarrow H^0(X - A, F) \xrightarrow{\partial} H_A^1(X, F) \dots$$

$$\dots H_A^i(X, F) \longrightarrow H^i(X, F) \longrightarrow H^i(X - A, F) \xrightarrow{\partial} H_A^{i+1}(X, F) \dots$$

This exact sequence shows that the cohomology group $H_A^i(X, F)$ plays the role of a relative cohomology group of $X \bmod X - A$, with coefficients in F . It is in this capacity that it was introduced naturally in the applications. By “sheafifying” (24) and (27), or by proceeding directly, one finds taking into account that the sheaf associated to $U \mapsto H^i(U, F)$ is zero if $i > 0$:

COROLLARY 2.10. *Under the conditions of 2.8, one has an exact sequence functorial in F :*

$$(24 \text{ bis}) \quad \dots \underline{H}_{Z'}^i(F) \longrightarrow \underline{H}_Z^i(F) \longrightarrow \underline{H}_{Z''}^i(F) \xrightarrow{\partial} \underline{H}_{Z'}^{i+1}(F) \dots$$

COROLLARY 2.11. *Let A be a closed subset of X , then one has an exact sequence functorial in F :*

$$(28) \quad 0 \longrightarrow \underline{H}_A^0(F) \longrightarrow F \longrightarrow f_*(F|_{X-A}) \xrightarrow{\partial} \underline{H}_A^1 \longrightarrow 0,$$

and canonical isomorphisms, for $i \geq 2$:

$$(29) \quad \underline{H}_A^i(F) = \underline{H}_{X-A}^{i-1}(F) = R^{i-1} f_*(F|_{X-A}),$$

where $f : (X - A) \rightarrow X$ is the inclusion.

This therefore defines $\underline{H}_A^0(F)$ and $\underline{H}_A^1(F)$ respectively as ker and coker of the canonical homomorphism

$$F \longrightarrow f_* f^*(F) = f_*(F|_{X-A}),$$

and the $\underline{H}_A^i(F)$ ($i \geq 2$) in terms of the derived functors of f_* .

COROLLARY 2.12. *Let F be an abelian sheaf on X . If F is flasque, then for every locally closed subset Z of X and every integer $i \neq 0$, one has $H_Z^i(X, F) = 0$, $\underline{H}_Z^i(F) = 0$. Conversely, if for every closed subset Z of X one has $H_Z^1(X, F) = 0$, then F is flasque.*

⁵N.D.E. see the previous note.

Suppose that F is flasque, then F induces a flasque sheaf on every open set, hence to prove $H_Z^i(X, F) = 0$ for $i > 0$, one may assume Z closed, and then the assertion follows from the exact sequence (27)⁶. One concludes from this for every locally closed Z , by “sheaffifying” i.e. applying 2.4, that $\underline{H}_Z^i(F) = 0$ for $i > 0$. Conversely, suppose $H_Z^1(X, F) = 0$ for every closed set Z , then the exact sequence (27)⁷ proves that for every such Z , $H^0(X, F) \rightarrow H^0(X - Z, F)$ is surjective, which means that F is flasque.

Combining 2.6 and 2.8, one will deduce from this:

PROPOSITION 2.13. *Let F be an abelian sheaf on X , Z a closed subset of X , $U = X - Z$, N an integer. The following conditions are equivalent:*

- (i) $\underline{H}_Z^i(F) = 0$ for $i \leq N$.
- (ii) For every open set V of X , considering the canonical homomorphism

$$H^i(V, F) \longrightarrow H^i(V \cap U, F),$$

this homomorphism is:

- a) bijective for $i < N$,
- b) injective for $i = N$.

(When $N > 0$, one may in (ii) confine oneself to requiring a)).

To prove (i) $\Rightarrow$ (ii), one is reduced, thanks to the local nature of the $\underline{H}_Z^i(F)$, to proving the

COROLLARY 2.14. *If condition 2.13 (i) is satisfied, then*

$$H^i(X, F) \longrightarrow H^i(U, F)$$

is bijective for $i < N$, injective for $i = N$.

Indeed, by virtue of the exact sequence (27), this also means $H_Z^i(X, F) = 0$ for $i \leq N$, and this relation is an immediate consequence of the spectral sequence (23 bis)⁸.

Conversely, the hypothesis 2.13 (ii) means that for every open set V of X , one has

$$H_{Z \cap V}^i(V, F|_V) = 0 \text{ for } i \leq N,$$

which implies 2.13 (i) thanks to 2.4. If moreover $N > 0$, hypothesis b) is superfluous. Indeed, if $N = 1$, hypothesis a) and (28) ensure the vanishing of $\underline{H}_Z^i(F) = 0$ for $i \leq N$. If $N > 1$, hypothesis a) for $i = N - 1 > 0$ and (29) ensure the vanishing of $\underline{H}_Z^i(F)$ for $i \leq N$.

Taking 2.11 into account this proves again 2.13 (i)...

REMARK. Let $Y \rightarrow X$ be a closed immersion, and suppose that locally it is of the form $\{0\} \times Y \subset R^n \times Y$. Suppose that F is a locally constant sheaf on x , then one finds

$$(30) \quad \underline{H}_Y^i(F) \simeq \begin{cases} 0 & \text{if } i \neq n \\ F \otimes \underline{T}_{Y,X} & \text{if } i = n, \text{ where } \underline{T}_{Y,X} \simeq \underline{H}_Y^n(\mathbf{Z}_X) \end{cases}$$

is an extension to X of a sheaf on Y locally isomorphic to $\mathbf{Z}_Y$, called “normal orientation sheaf of Y in X ”.

Using the spectral sequence (23 bis)⁹, one finds in this case:

$$(31) \quad H_Y^i(X, F) \simeq H^{i-n}(Y, F \otimes \underline{T}_{Y,X}),$$

and one recovers the “Gysin homomorphism”:

$$(32) \quad H^j(Y, F \otimes \underline{T}_{Y,X}) \longrightarrow H^{j+n}(X, F).$$

⁶N.D.E. the original reference was (2.9).

⁷N.D.E. the original reference was (2.9).

⁸N.D.E. see the previous note.

⁹N.D.E. the original reference was 2.6.

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